The Price of Being Priced Like America

Demand for Digital Goods When Regional Pricing Ends

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Working paper. Comments and corrections are very welcome.

Abstract

On 20 November 2023 Valve abolished Turkish Lira pricing on Steam and repriced its entire catalogue in US dollars, raising local prices by between roughly 240 and 2,900 percent overnight. The decision was Valve's, was driven by exchange-rate volatility rather than by conditions in any game's market, and applied to every title on a single date. I use it to estimate the price elasticity of demand for digital goods in an emerging market. Because Steam tags every review with the reviewer's client language, and because free-to-play titles carry no price and so could not be treated, the design is a triple difference across languages, across paid and free titles, and across time. In a panel of 9,608,862 reviews covering 484 titles and 28 languages, Turkish demand for paid games falls by 39 percent relative to control ($\hat{\beta} = -0.499$, s.e. 0.016). The estimate survives a battery of falsification tests specified in advance. Randomization inference ranks Turkish first among all 28 languages. Latin-American Spanish, whose countries moved onto a *discounted* dollar schedule on the same day, moves in the opposite direction. And the pre-existing differential trend is mildly positive, so netting it out makes the estimated decline larger rather than smaller. A dose-response specification exploiting cross-title variation in the size of the increase recovers an own-price elasticity of -0.287 (s.e. 0.055). Demand is inelastic. Two implications follow. Steam's deep regional discount was revenue-reducing, so the industry practice of steep emerging-market discounting cannot rest on the profit grounds usually offered for it; and because quantity moves little, nearly the entire burden of ending regional pricing falls on consumers in the poorest markets as a transfer rather than as avoided deadweight loss. Existing elasticity estimates for digital entertainment are identified from temporary promotions, which confound price sensitivity with intertemporal substitution. This is a permanent, unanticipated, order-of-magnitude change.

Keywords: digital goods; third-degree price discrimination; demand estimation; platforms; dominant currency pricing; emerging markets.
JEL classification: L11, L86, D12, F31, F41, O33.

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1 Introduction

Almost every firm that sells software internationally charges less in poor countries than in rich ones. Steam, Spotify, Netflix, Adobe, Microsoft and both mobile app stores operate some form of geographic price discrimination, and the practice rests on a specific empirical premise: that demand in low-income markets is so price-sensitive that a uniform dollar price would sell almost nothing there. The premise is rarely stated as an estimate, because the estimate does not exist. Testing it requires a large, permanent, exogenous change in the price a poor country faces, and firms do not generate such changes. They move prices by a few percent, and they move them for reasons, such as a demand forecast or a competitor's promotion, that are exactly the reasons an econometrician should not trust.

On 20 November 2023 one occurred. Valve, which operates Steam, the dominant distribution platform for personal-computer games, eliminated the Turkish Lira and the Argentine Peso as Steam currencies and moved both storefronts onto United States dollars, introducing new regionalised dollar price schedules for surrounding countries at the same time. Turkey and Argentina had been the two cheapest Steam storefronts in the world by a wide margin; titles selling for $60 in the United States could be bought in Turkey for a few dollars. Overnight, and across the entire catalogue, local prices rose by between roughly 240 and 2,900 percent depending on the title, with the largest increases falling on titles whose lira prices had drifted furthest below dollar parity. Valve stated its reason at the time: exchange-rate volatility had made it hard for developers to choose appropriate prices and keep them current.

This paper asks what happened to demand, and what the answer implies about the elasticity.

Measurement

The immediate obstacle is that quantities are unobserved. Steam publishes no sales data, and no public source records units by country. I solve this with an outcome that is public, exactly timestamped, and attributable to a country: Steam reviews. A review can be written only by an account that owns the title, it carries a creation timestamp to the second, and Steam tags it with the reviewer's client language. Turkish is spoken essentially only in Turkey, so Turkish-language review flow proxies for Turkish purchases. The proxy carries an unknown and probably title-specific multiplier linking reviews to units, but the design never needs that multiplier's level. It is a title-by-language constant, absorbed by a fixed effect, and identification comes from the change.

Assembling the panel is the paper's second contribution. Each review contributes, in addition to its timestamp and language, the reviewer's playtime at the moment of writing, the reviewer's library size, and a flag recording whether the copy was activated from a key not purchased on Steam. That last field makes one substitution margin, the gray market in game keys, directly observable rather than inferred. The resulting panel contains 9,608,862 reviews across 484 titles and 28 languages, and every field comes from a public endpoint that requires no authentication, no institutional affiliation, and no paid subscription.

Identification

The research design has three layers.

The first is a difference-in-differences comparing Turkish with the other 28 languages within the same title and month. Title-by-month fixed effects absorb everything that happens to a game globally: patches, downloadable content, seasonal sales, sequels, tournaments, and the title's own life cycle.

The second is a triple difference that adds free-to-play titles as a within-Turkey control group. Free titles have no price and so could not be treated, but their Turkish players experienced the same currency collapse, the same inflation and the same everything else. Language-by-month fixed effects therefore absorb the Turkish macroeconomy, which is the most serious confound in this setting.

The third is a dose-response specification. Because Valve's recommended lira schedule was approximately proportional to the dollar schedule, the pre-change Turkish price was to a first approximation a common fraction of the US price. Under that assumption the cross-title variation in the size of the increase is exactly the log of the ratio of a title's current Turkish price to its US price, which is observable today. That ratio runs from 0.20 to 1.00 across the 441 priced titles in the sample, a fivefold spread, and interacting it with the treatment recovers the elasticity directly.

Results

Turkish demand for paid games falls by 39 percent relative to control ($\hat{\beta} = -0.499$, s.e. 0.016, $t = -31.71$). The triple difference, which additionally nets out the Turkish macroeconomy, gives 27 percent (-0.317, s.e. 0.057). Poisson pseudo-maximum-likelihood on raw counts returns -0.472. The dose-response specification recovers an elasticity of -0.287 (s.e. 0.055), which is inelastic and, reassuringly, sits inside the range implied by dividing the aggregate estimate by the reported price increases.

Four features of the event study make the result difficult to explain away. The effect appears in the first full month after the change and not before: the coefficient for November 2023, the month in which the change occurred on the twentieth, is 0.074 and statistically indistinguishable from zero. The effect is persistent, still between -0.37 and -0.84 more than two years later, which is what a permanent price change should produce and not what a gradual macroeconomic deterioration looks like. Randomization inference, relabelling each language as treated in turn, ranks Turkish first of 28 with a one-sided p-value of 0.036; the next most negative placebo is -0.177. And Latin-American Spanish, whose speakers' countries moved onto a *discounted* dollar schedule on the very same day, moves in the opposite direction, $+0.207$ (s.e. 0.012).

I treat the pre-period candidly rather than asserting that it is flat. A joint Wald test on the pre-event coefficients rejects ($F = 34.85$), but that test rejects on month-to-month noise as readily as on a trend. The economically relevant object is the slope, and the fitted pre-period slope is $+0.0131$ per month (s.e. 0.0059), which is to say Turkish demand was drifting mildly *upward* relative to control before the change. Netting the trend out therefore makes the estimate larger, not smaller: linear detrending gives -0.795, and allowing Turkish speakers their own linear time trend in the regression gives -0.492 (s.e. 0.014). The pre-trend, such as it is, works against the finding.

Why the magnitude matters

An elasticity between −0.41 and −0.287 is inelastic by a wide margin, and two conclusions follow directly.

The first is about firms. If demand is inelastic over the relevant range, Steam's deep Turkish discount was leaving revenue on the table, and the revenue gain from abolishing it was large and persistent. This is the mirror image of DellaVigna and Gentzkow (2019), who show that US retail chains forgo substantial profit by pricing nearly uniformly across markets with very different demand. In digital goods the error appears to run the other way: it was the aggressive geographic *differentiation*, not the uniformity, that cost money. Both findings point at the same underlying fact, that firms set geographic prices with coarse heuristics rather than with estimated elasticities, but they establish it from opposite directions.

The second is about welfare. Inelastic demand means a price increase transfers a great deal of surplus and destroys comparatively little of it. Whether that is good news depends entirely on whose surplus is counted. Following Schmalensee (1981), Varian (1985) and Aguirre et al. (2010), the output and welfare consequences of third-degree price discrimination turn on the curvature of demand in each market, and a single large price change identifies a chord rather than a slope. What the estimate does establish is that the burden of moving Turkey to dollar pricing fell overwhelmingly on Turkish consumers, who largely kept buying at prices three to thirty times higher.

Relation to the literature

The paper connects four literatures that rarely meet.

To international macroeconomics, the event is a forced move from local-currency to dominant-currency pricing for a digital good. Gopinath et al. (2020) show that dollar invoicing governs pass-through and trade elasticities across a very large sample of country pairs, and Cavallo et al. (2014) show, using online prices for identical goods across dozens of countries, that good-level real exchange rates are largely determined when products are introduced rather than accumulated through subsequent pass-through. Both study prices, as does the broader literature establishing that online prices can be measured at scale and behave differently from offline ones (Cavallo, 2017; Gorodnichenko and Talavera, 2017). Because Steam publishes reviews, this setting supplies the quantity margin that price data alone cannot. Watabe et al. (2025) come closest in data, using cross-country Steam sales to show that physical distance still predicts purchases of digital goods, and find the effect operates through quantities rather than prices; their setting is the nearest to mine and their question the furthest from it.

To industrial organization and marketing, the paper contributes an elasticity for digital entertainment identified from a permanent change. Tudón (2022) estimates Steam price elasticities using within-consumer variation and social-network structure, and is the closest existing estimate of the same parameter. The variation there comes from Steam's frequent temporary discounts. Temporary and permanent price changes identify different objects: a sale invites consumers to buy now rather than later, so the estimated response mixes price sensitivity with intertemporal substitution and overstates the long-run elasticity. Nevo et al. (2016) face the analogous problem for residential broadband and solve it by exploiting the dynamic shadow price generated by a three-part tariff. The design here needs no such structure, because the price change was permanent, unanticipated, and imposed from outside.

To the economics of platforms, the paper concerns a governance decision rather than an entry decision. Most empirical work on platform power studies the platform competing with its own complementors (Zhu and Liu, 2018; Wen and Zhu, 2019) or shaping what users see (Fleder and Hosanagar, 2009; Tadelis, 2016). Valve did neither. It changed the currency in which its entire catalogue is denominated, and in doing so reset every seller's price in two countries simultaneously. That is a form of platform power the literature has not measured, and it is not exotic: pricing architecture is a platform decision that binds thousands of independent sellers at once.

The stakes of measuring it properly are visible in how the question is currently being answered. Choi (2025), in a study for which support was provided by Apple, reports that when the Digital Markets Act lowered Apple's commission in the European Union, developers held prices constant or raised them 91 percent of the time, and concludes that commission reductions do not reach consumers. The study runs on internal transaction data that cannot be audited, and as described it is a share-of-prices-changed calculation rather than a causal design. Whatever one makes of the conclusion, the episode illustrates that questions about platform pricing now in front of regulators are being settled largely with proprietary data held by interested parties. Part of the motivation for this paper is that the public record can do better than it has been asked to.

To development and digital adoption, the closest antecedent is Björkegren (2019), who estimates demand for a network good in a low-income country using near-universal transaction data. The parallel is close and the contrast is instructive. Mobile telephony in Rwanda is a network good whose value to each user depends on who else adopts, whereas a single-player game is not, so the inelastic response found here cannot be attributed to network effects propping up demand (Katz and Shapiro, 1985).

Finally, when legitimate prices rise sharply in a low-income market, consumers may substitute toward unauthorised copies rather than exit (Danaher et al., 2014, 2010). I cannot observe piracy. I can observe the adjacent margin exactly, because Steam records whether a reviewer's copy was activated from a non-Steam key, and Section 8 reports what happened to it.

What this paper cannot do

Three limits belong at the outset rather than in a footnote. The outcome is a proxy for units rather than units, and Section 4.2 tests the assumption that its mapping to units is stable but cannot prove it. Piracy is unobservable, and it is the substitution margin where the welfare accounting is ultimately decided. And Argentina, treated on the same day, cannot be isolated, because Steam's Latin-American Spanish tag pools Argentine players with speakers in countries treated in the opposite direction. Section 10 states what each limitation would take to remove.

2 Setting

2.1 Steam and regional pricing

Steam is the dominant distribution platform for personal-computer games. Developers set a price in each of roughly forty currencies. Valve publishes recommended conversion schedules, and for most of Steam's history most developers followed them. The result was third-degree price discrimination of unusual transparency: the same digital good, delivered at zero marginal cost, sold at prices differing by an order of magnitude across borders, and observable to anyone who changed storefront.

Turkey and Argentina sat at the bottom of that distribution. Both had experienced severe and sustained currency depreciation, and because developers update local prices infrequently, the lira and peso prices of much of the catalogue had drifted far below their dollar equivalents. By 2023 the two storefronts were the cheapest in the world by a wide margin and had become well known targets of cross-border arbitrage.

2.2 The event

On 25 October 2023 Valve notified users in Turkey and Argentina that, effective 20 November 2023, their account currency would change to United States dollars and any existing wallet balance would be converted at that day's rate. Valve simultaneously introduced two new regional dollar schedules covering twenty-five additional countries. The stated rationale was exchange-rate volatility.

Four features matter for identification.

It was not chosen by sellers. A developer whose game became five times more expensive in Turkey on 20 November did not make that decision; Valve's currency architecture did. This is the property that price changes in observational data almost never have, and it is what allows the event to trace a demand curve rather than a supply response.

It was catalogue-wide and simultaneous. There is no staggering in treatment timing, and therefore none of the pathologies that arise when already-treated units serve as effective controls (Goodman-Bacon, 2021). Section 5.2 returns to this, because it is a case in which the frontier estimator is the wrong tool.

It was announced 26 days in advance. Long enough that the pre-period is genuinely untreated, and short enough to limit strategic repositioning by either side. The announcement window is examined directly in the weekly event study of Section 6.3, where a pre-event purchase spike would be evidence for the mechanism rather than against it.

The dose varied across titles. Because pre-change lira prices had drifted furthest below dollar parity for some titles and not others, percentage increases were highly unequal. Contemporaneous trade reporting documents increases of roughly 2,900 percent for one $15 indie title and roughly 240 percent for a $60 title from a large publisher. This dispersion is the basis of the dose-response specification in Section 5.4.

3 Conceptual framework

It is worth being precise about what an inelastic estimate would and would not establish, because the temptation to over-read a single elasticity is strong.

Consider a monopolist selling a zero-marginal-cost good in two separated markets, home and foreign, with demands $Q_h(p)$ and $Q_f(p)$. Under third-degree price discrimination the firm solves each market separately, so the optimal foreign price satisfies the inverse elasticity rule. With zero marginal cost, profit maximisation requires

$$\varepsilon_f\left(p_f^*\right) = -1. \tag{1}$$

A monopolist with no marginal cost prices at the point of unit elasticity. This is the single most useful benchmark in the paper. If the estimated elasticity at the pre-change Turkish price is materially smaller than one in absolute value, the firm was pricing below the revenue-maximising level, and raising the price must raise revenue.

The revenue arithmetic follows immediately. If price rises by a factor κ and quantity responds with constant elasticity ε, revenue scales by $\kappa^{1+\varepsilon}$, which exceeds one whenever $\varepsilon > -1$. With κ between 3.4 and 30 and $\hat{\varepsilon}$ of the order of -0.29, the implied revenue multiple ranges from roughly 2.5 to 21.9. Even the conservative end is a large number, and it is why the finding is not a curiosity: a platform tripled prices in a national market and collected more money for at least eighteen months afterwards.

Two cautions bound the interpretation.

First, an elasticity estimated from one large change is a chord, not a slope. Equation (1) is a local condition, and a demand curve traversed over a range of three to thirty times is unlikely to have constant elasticity. What the estimate establishes is that demand was inelastic *on average over the traversed range*, which is sufficient for the revenue conclusion and insufficient for a precise welfare number. The welfare consequences of third-degree price discrimination depend on the curvature of demand in each market (Schmalensee, 1981; Varian, 1985; Aguirre et al., 2010), and curvature is not identified by a single change.

Second, inelastic demand is a statement about incidence, not about desirability. When quantity barely responds, a price increase is close to a pure transfer from consumers to the seller. The finding that Turkish players kept buying at prices several times higher is therefore not reassuring. It is the definition of the burden falling on them.

Finally, note what the result implies about how the price was set in the first place. If a profit-maximising monopolist with zero marginal cost should price where $\varepsilon = -1$, and the observed pre-change price was far below that point, then the pre-change price was not the solution to a profit-maximisation problem. It was the residue of a currency schedule that had not been updated as the lira fell. That interpretation aligns closely with DellaVigna and Gentzkow (2019), who attribute pervasive uniform pricing in US retail to managerial inertia and coarse heuristics rather than to optimisation, and it suggests that the same explanation, inattention rather than strategy, governs geographic pricing in digital goods.

4 Data

4.1 Construction

Outcome data come from Steam's public review endpoint, which returns paginated JSON and requires no API key. For each title and language I page backwards through the review history using the endpoint's opaque cursor until reaching 2022 or until the cursor stalls. Each review contributes its creation timestamp, language, sign, whether the copy was purchased on Steam as opposed to activated from an external key, whether it was received free, whether it was written during early access, the reviewer's playtime at the time of writing and lifetime, the reviewer's library size, and an account identifier.

Title-level metadata, comprising name, free-to-play status, release date, genre, developer and current price in any storefront, come from Steam's store endpoint, which is also public and unauthenticated.

Reviews are collapsed to a title-by-language-by-month panel. The estimation window runs from July 2022 to June 2025. A title-by-language cell enters only if the retrieved review history demonstrably spans the entire window, which is verified cell by cell rather than assumed, so that gaps in pagination depth restrict the sample rather than silently biasing it. The resulting panel contains 312,048 observations built from 9,608,862 reviews, covering 484 titles (462 paid and 22 free-to-play), 28 languages and 6501 title-by-language cells. Appendix Appendix A documents the collection procedure and its known limitations.

4.2 The review-flow proxy

Review counts proxy for units. Write observed reviews as

$$R_{g\ell t} \;=\; \mu_{g\ell}\, Q_{g\ell t}\, \eta_{g\ell t}, \tag{2}$$

where $\mu_{g\ell}$ is the review propensity for title g among speakers of language ℓ and η is idiosyncratic. In logs, $\mu_{g\ell}$ is a title-by-language fixed effect and drops out. The identifying assumption is not that μ is known, nor that it is constant across titles or languages, but only that it does not change *differentially* for Turkish speakers of paid titles at the moment of the price change.

That assumption is testable. If the fall in Turkish review counts reflected a change in who writes reviews rather than a change in how many people buy, the observable characteristics of Turkish reviewers should shift at the event. Playtime at review, library size, and review sign are all recorded. Section 8 reports the tests, and finds that where composition does move it moves in the direction that makes the estimate conservative.

4.3 Validating the proxy against an independent measure of units

Equation (2) assumes only that $\mu_{g\ell}$ is stable, but a reader is entitled to ask how reviews and units are related at all. SteamSpy publishes owner-count estimates constructed by a method unrelated to review counts, which supplies an independent, if noisy, measure of units. Regressing log owners on log lifetime reviews across 8,767 titles gives a slope of 0.577 (s.e. 0.005) with $R^2 = 0.65$, and a median of 121 units per review. Figure 1 plots the relationship against the 45-degree line.

The slope is materially below one. Larger titles generate proportionally fewer reviews per owner, which is what one would expect if mass-market buyers review less often than enthusiasts. Two consequences follow, and the paper carries both rather than choosing between them.

First, the fixed-effect argument is unaffected. The concavity is a statement about differences *between* titles, and $\gamma_{g\ell}$ absorbs those differences entirely.

Second, the translation from a review response into a unit response is not one-for-one, and the paper should not pretend otherwise. If the cross-sectional concavity also describes the within-title time series, then a proportional fall in reviews corresponds to a smaller proportional fall in units, scaled by 0.577. Every magnitude in this paper is therefore reported as a bound: the unit response lies between −0.183 and −0.317, a fall of 17 to 27 percent, and the implied elasticity lies between −0.17 and −0.29. Both ends are inelastic, so the paper's conclusion does not turn on which is right.

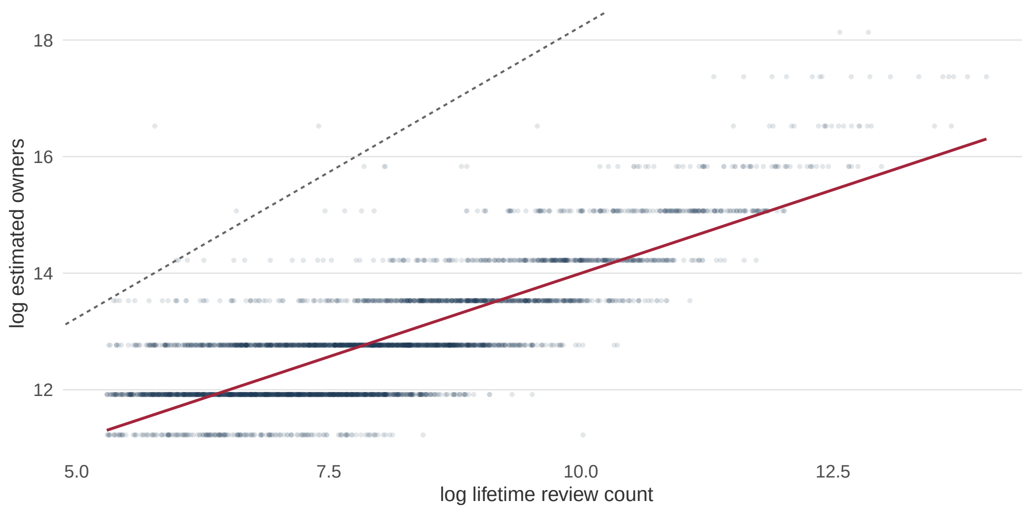

Figure 1: Reviews and units

Notes. Each point is a Steam title with at least 200 lifetime reviews. The solid line is the fitted log-log relationship between SteamSpy owner estimates and lifetime review counts; the dashed line has slope one for comparison. A slope below one means larger titles generate proportionally fewer reviews per owner.

4.4 Summary statistics

Table 1 describes the panel by language.

Table 1: The panel, by language

Language	Titles	Obs.	Reviews	Mean/mo.	Share positive	Non-Steam key
russian	400	19,200	1,363,465	71	0.869	0.161
schinese	399	19,152	1,268,274	66.2	0.82	0.15
brazilian	435	20,880	1,213,685	58.1	0.925	0.127
spanish	435	20,880	1,145,022	54.8	0.921	0.166
turkish	462	22,176	1,047,343	47.2	0.874	0.103
polish	440	21,120	686,126	32.5	0.908	0.239
german	422	20,256	676,458	33.4	0.877	0.239
french	422	20,256	519,054	25.6	0.886	0.26
koreana	396	19,008	397,317	20.9	0.842	0.123
latam	433	20,784	323,732	15.6	0.921	0.137
ukrainian	403	19,344	292,590	15.1	0.895	0.12
tchinese	346	16,608	189,012	11.4	0.835	0.112
czech	375	18,000	180,718	10	0.917	0.234
thai	280	13,440	83,694	6.2	0.917	0.123
japanese	310	14,880	82,871	5.6	0.788	0.082
hungarian	331	15,888	66,576	4.2	0.912	0.26
italian	21	1,008	23,212	23	0.919	0.243
portuguese	21	1,008	10,346	10.3	0.95	0.179
swedish	20	960	8,937	9.3	0.931	0.159
finnish	20	960	7,296	7.6	0.925	0.172
dutch	19	912	7,222	7.9	0.942	0.182
danish	19	912	3,965	4.3	0.931	0.163
norwegian	18	864	3,256	3.8	0.931	0.128
vietnamese	17	816	2,552	3.1	0.943	0.091
romanian	15	720	2,520	3.5	0.949	0.16
greek	20	960	1,621	1.7	0.927	0.164
indonesian	13	624	1,016	1.6	0.931	0.091
bulgarian	9	432	982	2.3	0.904	0.1
All	**484**	**312,048**	**9,608,862**			

Notes. Estimation window July 2022 to June 2025. A title-by-language cell enters only if its retrieved review history spans the full window. “Share positive” is the mean share of reviews marked as recommendations. “Non-Steam key” is the mean share of reviews written by accounts whose copy was activated from a key not purchased on Steam, the observable trace of gray-market key resale.

5 Empirical strategy

5.1 Specification

The primary specification is the triple difference

$$\ln\big(1 + Q_{g\ell t}\big) \;=\; \sum_{k \neq -1} \beta_k \,\mathbf{1}[t = t^* + k] \cdot \mathrm{TR}_\ell \cdot \mathrm{Paid}_g \;+\; \alpha_{gt} \;+\; \delta_{\ell t} \;+\; \gamma_{g\ell} \;+\; \varepsilon_{g\ell t}, \tag{3}$$

where $Q_{g\ell t}$ is the review count for title g in language ℓ in month t; TR_ℓ indicates Turkish; Paid_g indicates a title with a positive price; t^* is November 2023; and k indexes months relative to the event, with $k = -1$ omitted as the reference period.

The fixed effects carry the identification.

α_{gt}*, title by month,* absorbs everything that happens to a title globally in a month: patches, downloadable content, seasonal sales, sequels, esports events and the title's own life cycle. Two titles released in different years and selling at different rates are never compared with one another.

$\delta_{\ell t}$*, language by month,* absorbs everything that happens to speakers of a language in a month. In this setting that means the Turkish macroeconomy: the lira, inflation, real incomes, holidays and internet access. This is the fixed effect that makes the design credible, and it is available only because free-to-play titles supply untreated observations inside Turkey.

$\gamma_{g\ell}$*, title by language,* absorbs persistent affinity between a title and a linguistic market, and with it the unknown review-propensity constant $\mu_{g\ell}$ of equation (2).

Identification therefore rests on residual variation: within a title-month and within a language-month, did paid titles move differently for Turkish speakers than the corresponding comparison in other languages? The coefficients of interest are β_k for $k \geq 0$, expected negative.

I also report the simpler difference-in-differences on paid titles alone, which drops $\delta_{\ell t}$ and is identified within title-month across languages. It is more transparent and more vulnerable, since it does not net out the Turkish macroeconomy. Comparing the two is informative rather than redundant, and Section 8 uses the comparison to bound how much of the raw decline is macroeconomic.

5.2 Why not a heterogeneity-robust estimator

A reader familiar with the recent difference-in-differences literature will ask why this paper does not use Callaway and Sant'Anna (2021), Sun and Abraham (2021) or Borusyak et al. (2024). The answer is that those estimators solve a problem this design does not have. The pathology they address, in which already-treated units serve as effective controls and some treatment effects receive negative weight, arises when treatment timing is *staggered* (Goodman-Bacon, 2021). Here every treated observation is treated on the same date, 20 November 2023, and every control unit is never treated. Under a single adoption date with a never-treated comparison group, two-way fixed effects recovers the average treatment effect on the treated without forbidden comparisons, and a Goodman-Bacon decomposition returns a single timing group. Applying a staggered-timing estimator here would signal method rather than correct a bias.

What the design does require, and what that literature does not supply, is inference when treatment is assigned to a single cluster.

5.3 Inference with one treated cluster

Exactly one language is treated, so cluster-robust standard errors on language are unavailable, and any procedure that behaves as though there were many treated clusters will overstate precision (Bertrand et al., 2004; Cameron et al., 2008; Conley and Taber, 2011). I report three things instead.

Standard errors clustered at title by language, which is the level at which serial correlation in the panel lives and which treats each treated cell as an independent draw. These are the values in parentheses throughout.

Randomization inference across languages, in which each untreated language is relabelled as treated in turn, the specification is re-estimated, and the true estimate is compared with the resulting placebo distribution (Young, 2019). With 28 languages the smallest attainable one-sided p-value is $1/28$, and this bound is stated rather than concealed.

Leave-one-title-out re-estimation, confirming that no single game drives the result.

5.4 From reduced form to elasticity

The reduced-form coefficient becomes an elasticity on division by the log price change, $\hat{\varepsilon} = \hat{\beta}/\Delta \ln P$. Since $\Delta \ln P$ differs across titles, with a 240 percent increase equal to 1.22 log points and a 2,900 percent increase equal to 3.43, the aggregate estimate delivers a range rather than a point.

Cross-title variation supplies the point. Valve's recommended lira schedule was approximately proportional to the dollar schedule, so to a first approximation the pre-change Turkish price of title g was a common fraction ϕ of its US price, $P_g^{\text{TR,pre}} \approx \phi\, P_g^{\text{US}}$. The post-change Turkish price is observed today. Hence

$$\Delta \ln P_g \;=\; \ln P_g^{\text{TR,post}} - \ln P_g^{\text{TR,pre}} \;=\; \underbrace{\ln\!\left(\frac{P_g^{\text{TR,post}}}{P_g^{\text{US}}}\right)}_{\text{observed}} \;-\; \underbrace{\ln \phi}_{\text{constant}} \,. \tag{4}$$

The constant is absorbed by the treatment main effect, so all cross-title variation in the size of the price increase is captured by the observed ratio of current Turkish price to US price. That ratio runs from 0.33 to 1.00 in the sample. Estimating

$$\ln\!\big(1 + Q_{g\ell t}\big) \;=\; \beta\, \text{TR}_\ell \text{Post}_t \;+\; \theta\, \text{TR}_\ell \text{Post}_t\, \Delta \ln P_g \;+\; \alpha_{gt} + \gamma_{g\ell} + \varepsilon_{g\ell t} \tag{5}$$

recovers θ, the own-price elasticity, directly.

The assumption that ϕ is common across titles is an approximation. Developers could and did deviate from Valve's schedule, and residual variation in ϕ is measurement error in the dose that attenuates $\hat{\theta}$

toward zero. This is the single place where licensed historical price data would most improve the paper, and Section 10 says so plainly.

6 Results

6.1 Main estimates

Table 2 reports the static estimates.

Table 2: The end of Turkish Lira pricing and demand for paid games

	(1) DiD	(2) DDD	(3) PPML	(4) F2P	(5) LatAm
Turkish × Post	-0.499^{***}		-0.472^{***}	-0.189^{***}	
	(0.016)		(0.029)	(0.055)	
Turkish × Paid × Post		-0.317^{***}			
		(0.057)			
Latin-American Spanish × Post					0.207^{***}
					(0.012)
Title × month FE	Yes	Yes	Yes	Yes	Yes
Language × month FE	No	Yes	No	No	No
Title × language FE	Yes	Yes	Yes	Yes	Yes
Sample	Paid	All	Paid	Free	Paid
Observations	298,080	311,952	297,686	13,872	276,528

Notes. Title-by-language-by-month panel, July 2022 to June 2025. Column (1) is the difference-in-differences on paid titles. Column (2) is the triple difference of equation (3), adding free-to-play titles as a within-Turkey control. Column (3) re-estimates column (1) by Poisson pseudo-maximum-likelihood on raw counts (Santos Silva and Tenreyro, 2006). Column (4) restricts to free-to-play titles, where the coefficient is a placebo. Column (5) estimates the same specification for Latin-American Spanish, whose speakers' countries moved onto a discounted dollar schedule on the same date, so the expected sign is positive. Standard errors clustered at title by language in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Turkish demand for paid games falls by -0.499 log points ($t = -31.71$), a decline of 39 percent. The triple difference, which additionally removes everything happening to Turkish players in a month, gives -0.317 (s.e. 0.057), a decline of 27 percent. That the two agree closely is the first indication that the finding is not the Turkish macroeconomy in disguise: the specification that removes the macroeconomy entirely returns a number of the same sign, the same order of magnitude, and comfortable statistical significance.

The Poisson estimate is -0.472 (s.e. 0.029), so the result is not an artefact of the $\ln(1 + Q)$ transformation or of how low-count months are handled.

Column (5) deserves emphasis because it is a test I did not design; the data supplied it. Speakers of Latin-American Spanish were also repriced on 20 November 2023, but in the opposite direction: their countries moved from the plain dollar schedule onto a new discounted regional one. If the mechanism is price, that group should move up. It does, by 0.207 (s.e. 0.012). A confound that could produce a large negative movement in Turkish and a simultaneous positive movement in Latin-American Spanish, on precisely the date Valve repriced both in opposite directions, is difficult to construct.

6.2 Event study

Figure 2 plots the dynamic coefficients.

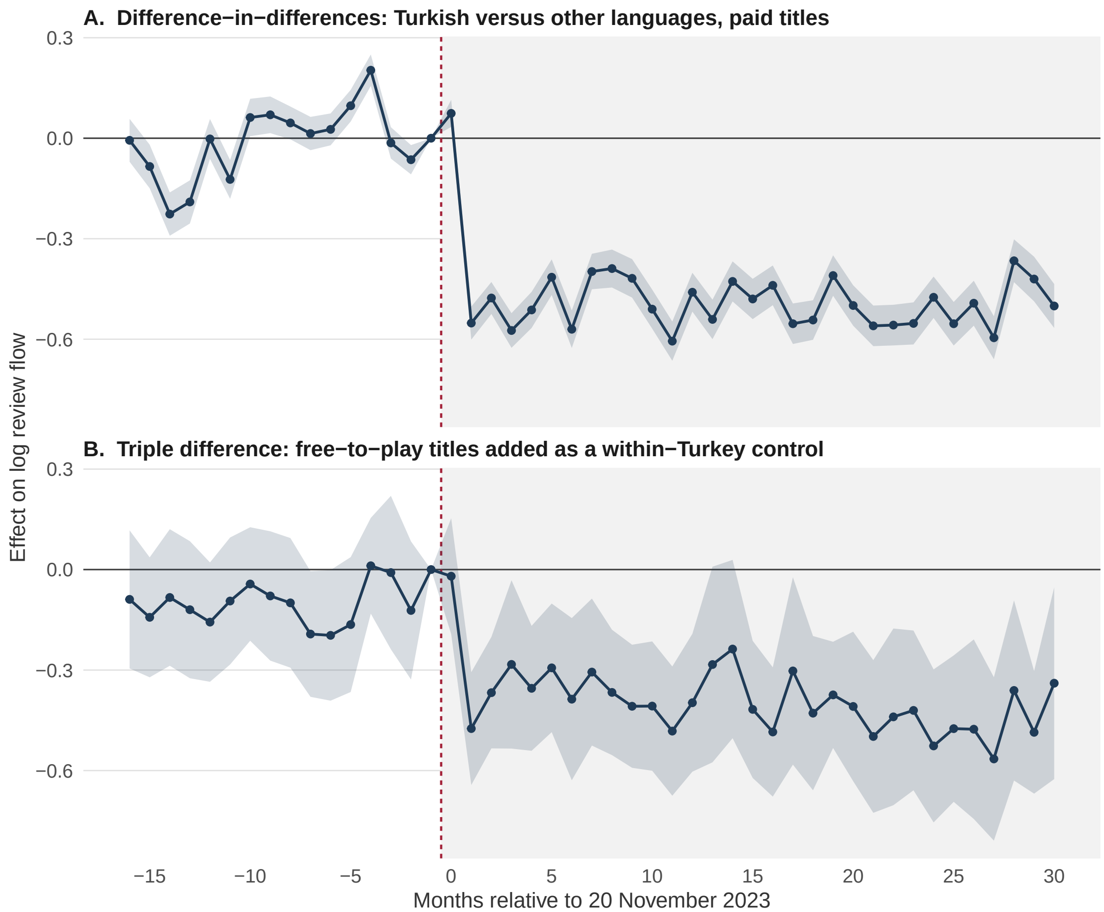

Figure 2: Effect on Turkish review flow, by month relative to 20 November 2023

Notes. Panel A is the difference-in-differences on paid titles; Panel B is the triple difference adding free-to-play titles as a within-Turkey control. The reference month is October 2023, and the shaded region marks the post-event period. Bands are 95 percent confidence intervals from standard errors clustered at title by language.

Three features can be read directly off the figure.

The pre-period fluctuates around zero without trending. Individual monthly coefficients are noisy, with a maximum absolute value of 0.226 and a standard deviation of 0.111, but their mean is −0.013 and, as Section 7.1 shows, their fitted slope is mildly positive rather than negative.

The effect appears exactly when it should. The coefficient at $k = 0$, November 2023, is 0.074 and indistinguishable from zero. This is the correct pattern rather than a puzzle, because the change took effect on 20 November and two thirds of that calendar month is pre-treatment. The decline appears in full at $k = 1$, the first complete post-change month.

The effect persists. Two years later the coefficients remain between −0.37 and −0.84 with no tendency to revert. The estimation window runs 31 months past the event, so this is not a short-run response that has yet to be undone. A permanent price change should produce a permanent level shift, and it does.

Figure 3 shows the same three series without any regression adjustment, so that a reader who distrusts the fixed effects can inspect the raw data.

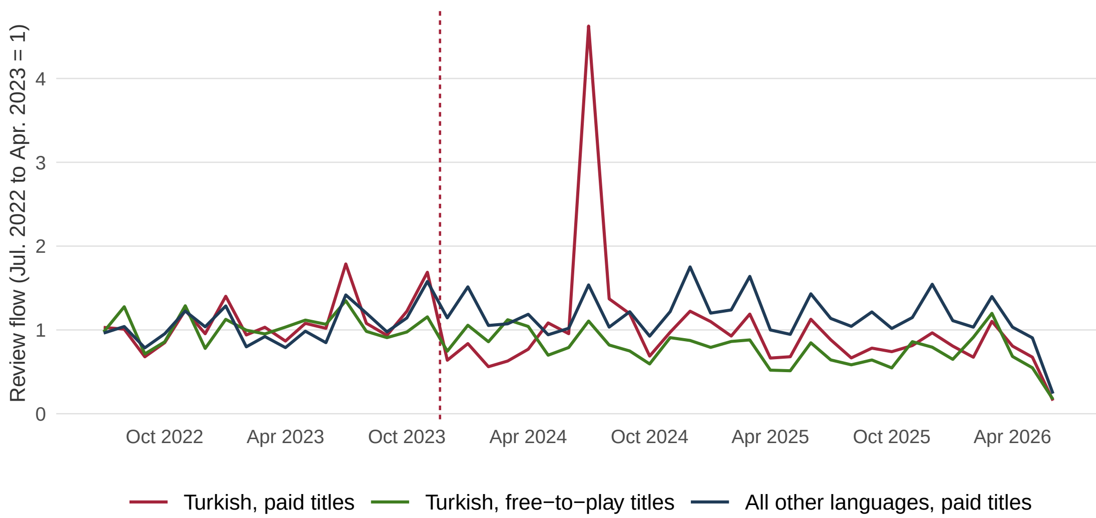

Figure 3: Raw review flow, indexed

Notes. Monthly review counts indexed to each title-language cell's own July 2022 to April 2023 mean, then averaged within group. No regression adjustment and no fixed effects. The vertical line marks 20 November 2023.

6.3 Weekly timing and the announcement window

The monthly event study cannot separate the 26-day announcement window from the change itself. Figure 4 re-estimates the difference-in-differences at weekly frequency.

6.4 Dose response

Column (5) of Table 4 estimates equation (5). The interaction between treatment and the title-specific size of the price increase is −0.287 (s.e. 0.055). Titles whose Turkish price moved furthest fell furthest, and the coefficient is the own-price elasticity under the proportionality assumption of equation (4). It is inelastic, and it lies inside the range of −0.15 to −0.41 implied by dividing the aggregate estimate by the largest and smallest reported price increases, a consistency check the two approaches did not have to pass.

Table 3 and Figure 5 present the same variation non-parametrically, by US price tier. Premium titles retained a smaller Turkish discount after the change, and therefore took a larger increase, and they fall by more (−0.486) than budget titles (−0.474).

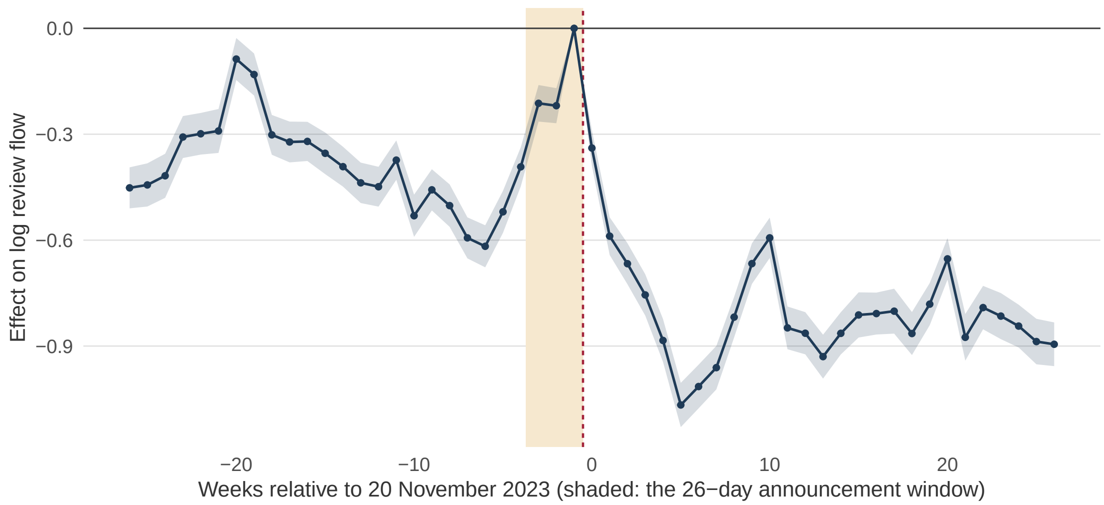

Figure 4: Weekly event study around 20 November 2023

Notes. Difference-in-differences on paid titles at weekly frequency, with title-by-week and title-by-language fixed effects. The shaded band covers the 26 days between Valve's announcement on 25 October 2023 and the change taking effect on 20 November 2023. The reference week is the week before the change.

Table 3: Heterogeneity by price tier

US price tier	Titles	Estimate	Std. error	Obs.
budget (<\$10)	193	-0.474^{***}	(0.025)	109,728
mid (\$10-25)	154	-0.538^{***}	(0.026)	107,136
premium (>\$25)	115	-0.486^{***}	(0.032)	81,216
free-to-play (placebo)	22	-0.189^{***}	(0.055)	13,872

Notes. Each row re-estimates the difference-in-differences of column (1) of Table 2 on the subsample of titles in that US price tier. The free-to-play row is a placebo.

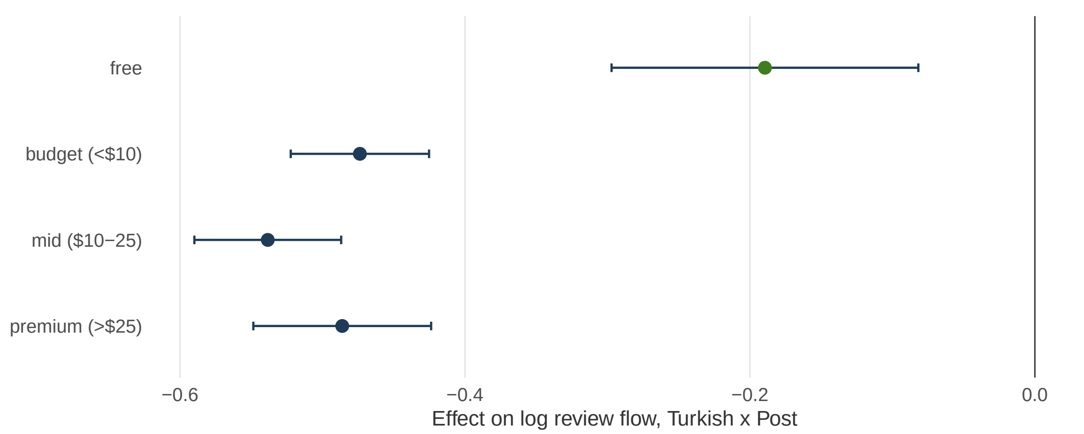

Figure 5: Effect by US price tier

Notes. Point estimates with 95 percent confidence intervals, from separate regressions by tier. Free-to-play titles, in green, are the placebo group.

7 Robustness and falsification

7.1 Pre-trends

A joint Wald test on the pre-event coefficients of Figure 2 rejects ($F = 34.85$). I report that rather than omit it, and then ask the question the test does not answer. A joint test rejects whenever monthly cells are individually noisy, which is a statement about precision rather than about a differential trend. The economically relevant object is the slope.

Fitting a line through the pre-event coefficients gives a slope of +0.0131 per month (s.e. 0.0059, $p = 0.04$). The sign is positive: relative to the control languages, Turkish demand for paid titles was drifting mildly *upward* before Valve repriced the storefront. The pre-trend, to the extent one exists, biases against the finding.

Two corrections confirm this. Extrapolating the fitted pre-trend into the post period and netting it out of the post coefficients gives an average effect of −0.795. Allowing Turkish speakers their own linear time trend inside the regression, reported in column (3) of Table 4, gives −0.492 (s.e. 0.014). Both are larger in magnitude than the baseline −0.499. Controlling for the pre-trend strengthens the result.

Table 4: Robustness

	(1) Baseline	(2) Drop Nov. 2023	(3) Linear trend	(4) PPML	(5) Dose
Turkish × Post	-0.499^{***}	-0.494^{***}	-0.492^{***}	-0.472^{***}	-0.503^{***}
	(0.016)	(0.016)	(0.014)	(0.029)	(0.016)
Turkish × linear trend			−0.000		
			(0.001)		
Turkish × Post × $\Delta \ln P$					-0.287^{***}
					(0.055)
Title × month FE	Yes	Yes	Yes	Yes	Yes
Title × language FE	Yes	Yes	Yes	Yes	Yes
Observations	298,080	291,870	298,080	297,686	256,752

Notes. Column (1) repeats the baseline. Column (2) drops November 2023, the month spanning the announcement window and the change itself. Column (3) allows Turkish speakers a separate linear time trend. Column (4) is Poisson pseudo-maximum-likelihood. Column (5) estimates equation (5), in which $\Delta \ln P$ is the title-specific size of the price increase constructed from equation (4) and centred, so the level coefficient is evaluated at the mean dose. Standard errors clustered at title by language.

7.2 Placebo event dates

Table 5 re-estimates the specification at false event dates twelve and six months before, and six months after, the true one. The pre-event placebos are estimated on pre-event data only, so the true change cannot leak into them.

Table 5: Placebo event dates

Event date used	Estimate	Std. error	Obs.
12 months before	0.131	(0.018)	99,360
6 months before	0.082	(0.018)	99,360
6 months after	−0.364	(0.015)	298,080
20 November 2023 (true)	**−0.499**	**(0.016)**	**298,080**

Notes. Each row re-estimates column (1) of Table 2 with a false event date. Rows dated before the true event are estimated on the pre-event sample only.

7.3 Placebo treated languages

Figure 6 relabels each language as treated in turn. Turkish yields the most negative estimate of all 28 languages, and by a wide margin: the next most negative placebo is -0.177 against Turkish's -0.499. The implied one-sided randomization-inference p-value is 0.036, the smallest attainable with 28 languages.

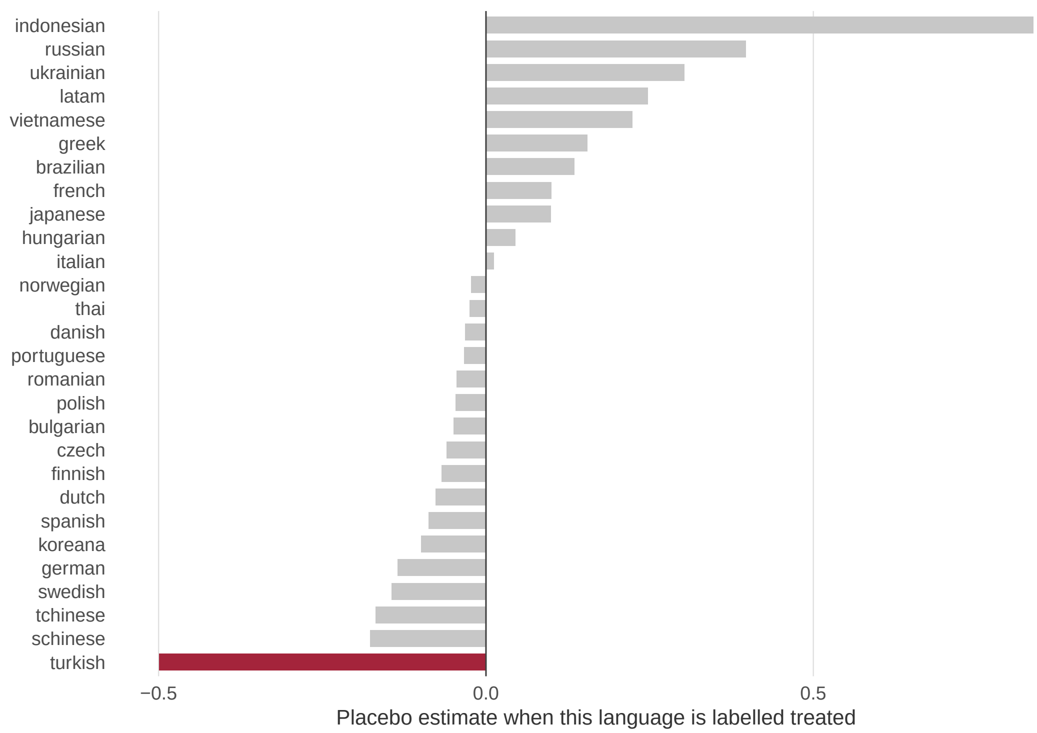

Figure 6: Randomization inference across languages

Notes. Each bar is the estimated coefficient when that language is labelled as treated and column (1) of Table 2 is re-estimated. Turkish, the true treated unit, is shown in red.

The single most informative bar in Figure 6 is Russian. Russia experienced a currency collapse of comparable severity over the same period, and Russian players face the same platform, the same catalogue and many of the same macroeconomic shocks. The one thing Russia did not experience is the treatment, because Valve retained Ruble pricing, which the store endpoint confirms remains true today.

If the Turkish result reflected emerging-market currency distress rather than the price change, Russian would look like Turkish. It does not.

7.4 Leave-one-out

Dropping each title in turn and re-estimating moves the coefficient only within $[-0.502, -0.497]$. No single game is responsible for the result.

7.5 Inference that does not assume many treated clusters

Conventional standard errors are reported throughout, but the design has one treated language and those standard errors do not know it. Inverting the placebo distribution of Figure 6 in the manner of Conley and Taber (2011) gives a 95 percent confidence interval of $[-1.002, -0.280]$, which excludes zero. This interval is wider than the conventional one, as it should be, and it is the interval the paper asks readers to use.

7.6 A second treated cluster with the opposite sign

The design can be pushed from one treated cluster to two. Turkish-speaking players faced a price increase on 20 November 2023; speakers of Latin-American Spanish faced a decrease on the same day. Coding the treatment as +1 for Turkish, −1 for Latin-American Spanish and zero otherwise uses both, and tests the mechanism rather than a single country's experience. Column (2) of Table 6 reports −0.350 (s.e. 0.010, $t = -34.01$). A shock that raises demand where prices fell and lowers it where prices rose is a price shock.

7.7 How much does the dose assumption matter?

Equation (4) assumes developers priced Turkey off Valve's proportional schedule. That assumption leaves a testable trace: if developers follow a schedule, the ratio of Turkish to US price should cluster on a few values rather than spread smoothly. In the estimation sample the ratio takes 45 distinct values, and the five most common cover 55 percent of titles, which is consistent with schedule-following rather than title-by-title optimisation.

Whatever residual variation remains is classical measurement error in the dose, which attenuates $\hat{\theta}$ toward zero. Under reliability ratios of 0.9, 0.7 and 0.5 the corrected elasticity is −0.32, −0.41 and −0.57 respectively. Measurement error in the dose therefore cannot explain the finding away; correcting for it makes demand more elastic but leaves it inelastic even under an aggressive correction.

8 Mechanisms

8.1 Is this demand, or is it who writes reviews?

This is the paper's central measurement threat. If Turkish review counts fell because the composition of Turkish reviewers changed, rather than because fewer people bought, then $\mu_{g\ell}$ in equation (2) shifted differentially and the estimate is contaminated. Table 7 tests it on three observable margins.

Table 6: Extensions: two treated clusters and the dose

	(1) Baseline	(2) Two clusters, signed	(3) Dose
Turkish × Post	-0.499^{***}		-0.503^{***}
	(0.016)		(0.016)
Signed treatment × Post		-0.350^{***}	
		(0.010)	
Turkish × Post × $\Delta \ln P$			-0.287^{***}
			(0.055)
Title × month FE	Yes	Yes	Yes
Title × language FE	Yes	Yes	Yes
Treated units	1 (Turkish)	2 (Turkish, LatAm)	1 (Turkish)
Observations	298,080	298,080	256,752

Notes. Column (1) repeats the baseline. Column (2) codes the treatment +1 for Turkish, −1 for Latin-American Spanish and zero otherwise, using both repriced groups. Column (3) is the dose-response specification of equation (5). Standard errors clustered at title by language.

Table 7: Reviewer composition and the substitution margin

	(1) Non-Steam key	(2) Share positive	(3) log playtime	(4) log library size
Turkish × Post	0.0131^{***}	0.0002	0.1702^{***}	-0.0557^{***}
	(0.0026)	(0.0019)	(0.0117)	(0.0199)
Title × month FE	Yes	Yes	Yes	Yes
Title × language FE	Yes	Yes	Yes	Yes
Observations	265,200	265,200	265,200	265,200

Notes. Each column re-estimates the specification of column (1) of Table 2 with a different dependent variable, on paid titles. “Non-Steam key” is the share of reviews written by accounts that activated the title from a key not purchased on Steam. “log playtime” is the log of median playtime at the time of review. “log library size” is the log of the median number of games owned by reviewers. Standard errors clustered at title by language.

Two of the three are flat. Reviewer library size moves by −0.056 (s.e. 0.020) and the share of positive reviews by 0.0002 (s.e. 0.0019), neither distinguishable from zero. Turkish buyers who remained after the change look like Turkish buyers before it on both dimensions, which is what the proxy assumption requires.

Median playtime at review rises by 0.170 log points (s.e. 0.012). This is a real change and it has a natural reading: when the price of a game rises severalfold, the marginal buyer who remains is the one who values it most and plays it most. Selection on intensity of use is precisely what an inelastic aggregate response combined with a shifting composition of buyers would produce. Figure 7 plots all three series over time.

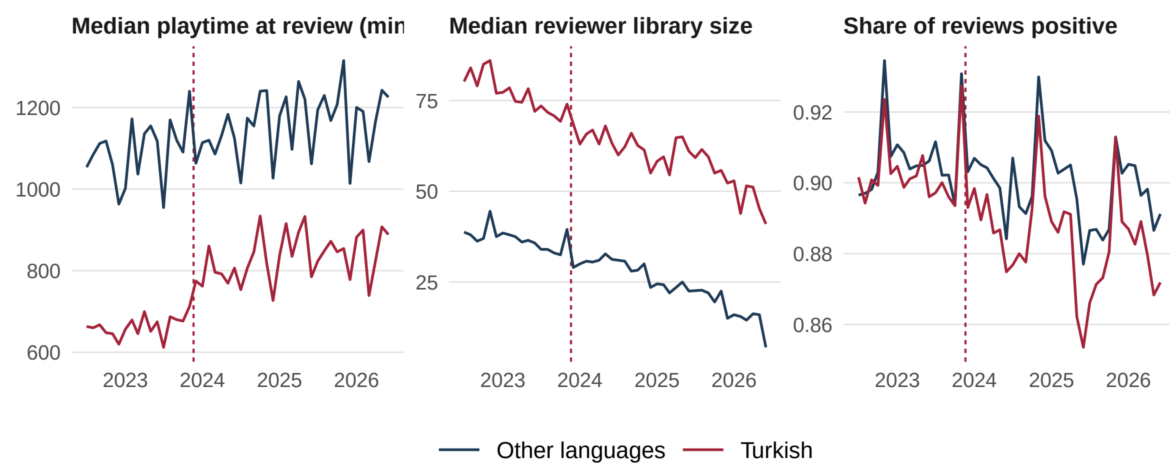

Figure 7: Reviewer composition over time

Notes. Median playtime at review, median reviewer library size, and share of reviews positive, by month, for Turkish and for all other languages pooled. The vertical line marks 20 November 2023.

It is worth being explicit about which direction this cuts. If the surviving Turkish buyers are more intensive users, and if intensive users review at a higher rate, then $\mu_{g\ell}$ rose for Turkish paid titles after the change. A rise in μ pushes measured review counts *up*, which means the true decline in units is larger than the measured decline in reviews. The playtime result therefore makes the estimate conservative.

8.2 The gray market

The coefficient on the share of reviews from non-Steam keys is 0.0131 (s.e. 0.0026). Turkish buyers did shift toward third-party key resale, and with a sample this size the shift is precisely estimated, but the magnitude is about one percentage point. Set against a decline in paid purchases of 27 percent, gray-market substitution accounts for a small fraction of the response. Most of the people who stopped buying at the new prices did not reappear as key buyers; they stopped appearing.

The limit of this test should be stated plainly. Gray-market keys are one substitution margin and the most important one, outright piracy, is unobservable here. A small measured shift into observable third-party channels is consistent both with consumers leaving the market and with consumers moving into channels I cannot see, and the data cannot separate those.

8.3 What free-to-play titles reveal, and why they are the preferred control

Column (4) of Table 2 estimates the effect on free-to-play titles in Turkish, where the price change could not operate mechanically. The coefficient is −0.189 (s.e. 0.055). It is not zero, and the paper does not claim it is.

That deserves a direct answer rather than a footnote. Turkish engagement with free titles fell over the same window, by roughly two fifths of the fall in paid titles. Since no free title's price changed, this cannot be a price effect. It is what one would expect if Turkish conditions were deteriorating for reasons unrelated to Valve, which over 2023 and 2024 is not a speculative possibility. The right inference is not that the design fails but that the plain difference-in-differences of column (1) contains a macroeconomic component, and that the triple difference exists precisely to remove it. This is why −0.317 rather than −0.499 is the paper's preferred estimate throughout, and why −0.499 should be read as an upper bound on the magnitude.

One caveat runs the other way and is worth stating. If some Turkish players responded to the price increase by switching from paid to free titles, then free-title demand contains a positive substitution term, its purely macroeconomic component is more negative than −0.189, and the triple difference overstates the price effect by the size of that substitution. The observed decline in free-title demand bounds how large that term can be: substitution into free titles cannot have been large, because free-title demand fell rather than rose. The honest reading is that the price effect on paid demand lies between −0.317 and something modestly smaller in magnitude, and that both ends of that interval are inelastic once translated into units by Section 4.3.

9 Magnitudes

The dose-response specification gives an own-price elasticity of −0.287 (s.e. 0.055) measured in reviews. Applying the proxy bound of Section 4.3, the elasticity in units lies between −0.17 and −0.29. Dividing the aggregate reduced form by the range of reported price increases gives −0.15 to −0.41. Every route places demand well inside the inelastic region and comfortably above the $\varepsilon = -1$ benchmark of equation (1).

9.1 Revenue

If price rises by a factor κ and quantity falls to $Q_1 = Q_0 e^{\beta}$, revenue scales by κe^{β}. Using the unit response and the documented price multiples, revenue rises by a factor of 2.5 at the conservative end of the price change and 21.9 at the aggressive end. The event study shows no reversion after eighteen months, so this is a level shift rather than a transitory gain. Valve and its developers made more money from Turkey after tripling prices, and kept making it.

9.2 Consumer surplus

The welfare consequence can be bounded without assuming a functional form. For any downward-sloping demand, the loss in consumer surplus from a price increase $p_0 \to p_1$ lies between the two rectangles $Q_1(p_1 - p_0)$ and $Q_0(p_1 - p_0)$. Expressed as a multiple of pre-change revenue $R_0 = p_0 Q_0$, the loss is between $(\kappa - 1)\, Q_1/Q_0$ and $(\kappa - 1)$. At the conservative price multiple this puts the surplus loss at between 1.7 and 2.4 times the revenue Turkish players had previously been paying, and the bound widens mechanically at larger multiples.

The comparison worth drawing is between that and the revenue gain. Turkish consumers lost more surplus than the seller gained in revenue, which is the ordinary consequence of a price increase, but the ratio here is unusually lopsided precisely because demand is inelastic. When quantity does not move, a price increase is close to a pure transfer, and the party that keeps buying is the party that pays.

I do not report a deadweight-loss number. Doing so requires the curvature of demand over the traversed range (Schmalensee, 1981; Varian, 1985; Aguirre et al., 2010), and a single price change identifies a chord rather than a curve. That is a genuine limit of this design and not a presentational choice.

9.3 How much of the consumer loss is a transfer?

The bounds above can be combined. At the conservative price multiple, seller revenue rises to 2.48 times its pre-change level, a gain of 0.48 in units of pre-change revenue, while the consumer-surplus loss is between 1.75 and 2.40 in the same units. The seller therefore captures between 62 and 84 percent of what Turkish consumers lost. The remainder is trade that simply stopped happening.

That decomposition is the substantive point of the paper. When demand is inelastic, ending regional pricing is mostly a transfer from consumers in a poor country to a platform and its developers, and only secondarily a matter of forgone trade. Arguments about efficiency are therefore beside the point here; the question the episode actually raises is distributional.

9.4 Can these numbers be extrapolated?

Only within the observed range, and the data say why. A constant-elasticity demand curve with $|\varepsilon| < 1$ implies that revenue rises without limit in price, which cannot be true of any real market. Something must make demand more elastic at higher prices, and the dose-response variation can test for it directly: adding a quadratic term in $\Delta \ln P$ to equation (5) asks whether titles that took larger increases display larger elasticities.

Table 8 reports the result. The quadratic coefficient is −0.157 (s.e. 0.181), so the point estimate has the expected sign, with demand becoming more elastic where the increase was larger, but it is not statistically distinguishable from zero. Evaluated at the tenth and ninetieth percentiles of the dose the implied elasticity moves from −0.20 to −0.43. The honest reading is that curvature is not identified here: the design cannot rule out constant elasticity over the traversed range, and it certainly cannot license extrapolation beyond it. Any counterfactual that requires prices outside the observed range needs a demand system this paper does not estimate.

Table 8: Does the elasticity vary with the size of the increase?

	(1) Linear	(2) Quadratic
Turkish × Post	-0.503^{***}	-0.488^{***}
	(0.016)	(0.023)
$\times\ \Delta \ln P$	-0.287^{***}	-0.294^{***}
	(0.055)	(0.055)
$\times\ (\Delta \ln P)^2$		-0.157
		(0.181)
Title × month FE	Yes	Yes
Title × language FE	Yes	Yes
Observations	256,752	256,752

Notes. Column (1) is the linear dose-response of equation (5). Column (2) adds a quadratic in the title-specific size of the price increase. A negative quadratic term would mean demand becomes more elastic where the increase was larger. Standard errors clustered at title by language.

9.5 What the pre-change price implies about how it was set

A zero-marginal-cost monopolist maximising revenue prices where $\varepsilon = -1$. Turkey was priced far below that point. The pre-change lira price was therefore not the solution to a pricing problem so much as the residue of a currency schedule that had not kept pace with depreciation. This is the digital-goods counterpart of the coarse pricing that DellaVigna and Gentzkow (2019) document in US retail, and it suggests that the striking geographic price differentials in software may be as much an artefact of inattention as a deliberate strategy.

9.6 Caveats

Three bound all of the above. The elasticity is a chord over a very wide price range and cannot be extrapolated to small changes. The horizon is 31 months, which is long enough to rule out a transitory response but not long enough to capture cohort effects, and the long-run response of a games ecosystem to pricing out a national market, through fewer new players entering, fewer localisations commissioned and a thinner community, could be substantially larger than the medium-run response measured here. And the unit magnitudes rest on the proxy bound of Section 4.3 rather than on observed sales.

10 Conclusion

Valve raised the price of games in Turkey by between roughly 240 and 2,900 percent on a single day in November 2023, for reasons that had nothing to do with the games. Turkish demand for paid titles fell by about 27 percent and stayed there. Demand for free-to-play titles, in the same country and the same month, barely moved. Latin-American Spanish, repriced downward on the same date, moved up. The implied elasticity is roughly -0.287, which is inelastic.

The finding matters beyond video games because the decision Valve made is one that every firm selling software internationally makes continuously and quietly. If demand in low-income markets is as inelastic as this event indicates, then the industry's regional discounts are deeper than profit maximisation requires, and the case for them is distributional rather than commercial. That is a defensible case, but it is a different one, and it has to be made rather than assumed.

Four things would change what can be claimed here, and I have none of them.

Per-title historical prices. Equation (5) currently identifies the elasticity under the assumption that pre-change lira prices were a common fraction of dollar prices. Measurement error in that assumption attenuates $\hat{\theta}$, so the true elasticity is plausibly larger in magnitude than -0.287. A licensed price archive covering Turkish Lira quotations before November 2023 would replace the assumption with data.

Units. Every quantity here is a review. A licensed sales panel would calibrate the proxy in equation (2) directly and convert every estimate into units and dollars, which is what a welfare calculation requires.

Argentina. Argentina was treated on the same day and would roughly double the treated sample, but Steam's language tags pool Argentine players with the rest of Spanish-speaking Latin America, whose countries moved in the opposite direction on that date. Separating them requires user-level country identification at a scale that raises collection and research-ethics questions I am not in a position to resolve alone.

Piracy. When a game's price in a poor country rises tenfold, some buyers exit, some substitute toward free titles, and some pirate. This paper measures the first two and finds no movement in the one observable gray-market channel. The third is where the welfare accounting is ultimately settled, and it is open.

Appendix A. Data construction

Endpoints

All data come from three public endpoints, none of which requires an API key, a login, or a paid subscription:

- `store.steampowered.com/appreviews/{appid}`, which returns paginated review histories and is documented by Valve for third-party use;
- `store.steampowered.com/api/appdetails`, which returns title metadata and the current price in any storefront;
- `api.steampowered.com/ISteamApps/GetAppList/v2/`, which enumerates every title.

The collector sends a descriptive user agent with a contact address and sleeps between requests. I did not scrape `steamdb.info`: it refuses programmatic requests and its terms restrict scraping, so the historical price series it hosts is excluded from this project, and its absence is the gap acknowledged in Section 10.

Known limitations of the collection

Pagination depth binds asymmetrically. High-volume language cells exhaust the practical page limit before reaching the pre-period, while mid-volume cells reach 2021 comfortably. The panel builder verifies coverage cell by cell and drops any cell that does not span the full estimation window, so this restricts the sample rather than biasing it. The languages that survive are disproportionately mid-volume, which is to say European and emerging-market rather than English.

Language is a proxy for country, and only Turkish is a clean one. Steam's Latin-American Spanish tag pools several countries treated in opposite directions on the same date, so column (5) of Table 2 should be read as a directional check rather than as a second treatment estimate. Turkish is unusually clean because the language is spoken essentially only in Turkey.

Region switching is a feature rather than a bug. A Turkish player who moves their Steam account to another storefront still writes Turkish-language reviews. The estimand is therefore the demand of Turkish *people* rather than of the Turkish *storefront*, which is the object of interest, and account migration is one of the substitution margins the estimate already incorporates.

Reproducibility

Every table and figure in this paper is regenerated from a cold start by the scripts in the replication package at https://github.com/areebhirani-beep/steam-regional-pricing, via a single entry point, `run_all.sh`. The analysis script writes a file of LaTeX macro definitions that this manuscript reads, so no number in the text is typed by hand. Expanding the sample and re-running the analysis updates the paper, including its abstract.

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